date : 2026-10-05,18:26

Reference

youtube video

OpenCourseWare - Mathematics for Computer Science - 1. introduction and proofs

Handwritten note

Math for CS - 1 introduction and proofs

proofs - a method for ascertaining the truth

↓ verifying / establish

many ways of ascertaining the truth

- expectation & observation
- sampling & counter example
- judge, jury
- religion
- conviction

A math proof is verification of a proposition by a chain of logical detection from a set of axioms

Proposition

Def: a statement that is either T or F

Ex: $2+3=5 \rightarrow T$

$\forall n \in \mathbb{N}$, n^2+n+41 = prime number

"for all" ↑ "natural number" ↑ $\{0, 1, 2, \infty\}$

↓ Predicate: proposition whose T depend on the value of variable (n)

↑ quantifier

n	n^2+n+41	prime
0	41	✓
⋮	⋮	⋮
40	1681	✓

T for first 40 ⋮

39 1681 $\times (91^2)$

40

Ex: $a^4 + b^4 + c^4 = d^4$ has no positive integer solution

can be prove wrong by

$a = 95,800$
$b = 217,519$
$c = 414,560$
$d = 422,481$

correct true proposition $\Rightarrow \exists\, a,b,c,d \in \mathbb{N}^+,\ \underbrace{a^4+b^4+c^4=d^4}_{\text{Predicate}}$

"these exist" $\{1,3,\dots\}$

Ex $313\,(x^3+y^3)=z^3$ has no positive integer solution

↳ false

don't try a few cases and if it work doesn't mean its done

Ex The regions in any map can be colored in 4 colors - so that a adjcent regions have different colors

$\forall n \in \mathbb{Z},\ n \geq 3? \rightarrow n^2 \geq 4$

↑ integer $\{0,1,-1,\dots\}$ ↖ implie

→ def: An implication $p \rightarrow q$ is true if p is f or q is T

Truth table

p	q	$p \rightarrow q$
T	T	T
T	F	F


F T
F F
T

- Ex: $\forall n \in \mathbb{Z}$, $n \geq 3 \iff n^2 \geq 4$
iff

False $n = -3$ $\iff = \Leftarrow, \Rightarrow$

what is proposistion → not question

---

axioms

Def: a proposition that is "assume" to be true
there is no proof just assume

Ex: If $a=b$ & $b=c$, then $a=c$

Euclidean geometry, given a line L, a point p not on L, there is only one line through p parallel to L

spherical geometry, " " " there is no line

hyperbolic geometry, " " " " There are infintely many

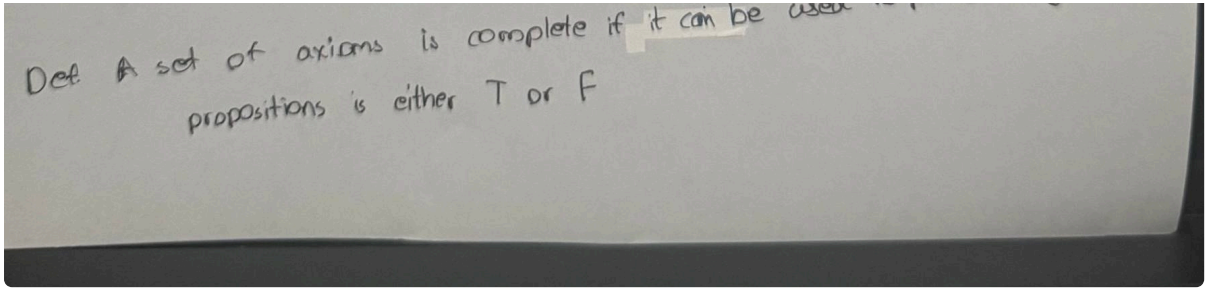
* axioms can be contradictory in different context
That is ok, state based on field

Axioms should be

1. consistent
2. coroplete

Def: A set of axioms is consistent if no proper can be proved to be both T and F

to prove every



book reading

1 Propositions

Definition : A proposition is a statement that is either true or false.

For example, both of the following statements are propositions. The first is true and the second is false.

Proposition 1.0.1 : 2 + 3 = 5.
Proposition 1.0.2 : 1 + 1 = 3

1.1 Compound Propositions

1.1.1 NOT, AND, and OR

A	NOT (A)
T	F
F	T

P	Q	P AND Q	P OR A	P XOR Q
T	T	T	T	F
T	F	F	T	T
F	T	F	T	T

P	Q	P AND Q	P OR A	P XOR Q
F	F	F	F	F

1.1.2 IMPLIES

An implication is true exactly when the if-part is false or the then-part is true.

P	Q	P IMPLIES Q	
T	**T**	**T**	(tt)
T	**F**	**F**	(tf)
F	**T**	**T**	(ft)
F	**F**	**T**	(ff)

1.1.3 IFF

P	Q	P IFF Q
T	**T**	**T**
T	**F**	**F**
F	**T**	**F**
F	**F**	**T**

1.1.4 Notation

English	Symbolic Notation
$\text{NOT}(P)$	$\neg P$ (alternatively, $\overline{P}$)
P AND Q	$P \wedge Q$
P OR Q	$P \vee Q$
P IMPLIES Q	$P \longrightarrow Q$
if P then Q	$P \longrightarrow Q$
P IFF Q	$P \longleftrightarrow Q$

1.1.5 Logically Equivalent Implications

P	Q	Proposition (P→Q)	Contrapositive (¬Q→¬P)	Converse (Q→P)	Inverse (¬P→¬Q)
T	**T**	T	T	T	T
T	**F**	F	F	T	T
F	**T**	T	T	F	F
F	**F**	T	T	T	T

an implication and its converse together are equivalent to an iff statement.

P	Q	(P↔Q)	P→Q And Q→P
T	**T**	T	T
T	**F**	F	F

P	Q	(P↔Q)	P→Q And Q→P
F	**T**	F	F
F	**F**	T	T

1.2 Propositional Logic in Computer Programs

Here is the step-by-step breakdown using standard boolean algebra / logic laws:

1. **Start with the expression:**

$$A \ (A \ B)$$

2. **Apply the Distributive Law:**

$$(A \ A) \ (A \ B)$$

3. **Apply the Complement Law / Negation Law ($A \ A = \text{True}$):**

$$\text{True} \ (A \ B)$$

4. **Apply the Identity Law ($\text{True} \ X = X$):**

$$A \ B$$

In summary, while it simplifies directly using the **Absorption/Redundancy Law** ($A \ (A \ B) = A \ B$), it is proved via the **Distributive Law**, **Complement Law**, and **Identity Law**

1.3 Predicates and Quantifiers

1.3.1 Propositions with Infinitely Many Cases

Proposition 1.3.1. For every nonnegative integer, n, the value of $n^2 + n + 41$ is prime.

there are many examples of propositions that seem to be true when you check a few cases (or even many), but which turn out to be false.

> The key to remember is that you can’t check a claim about an infinite set by checking a finite set of its elements, no matter how large the finite set

it can rewrite as

$$\forall n \in \mathbb{N}. (n) \text{ is prime.}$$

Symbol	Name / Meaning	Explanation
$\forall$	**Universal Quantifier**	Read as "for all" or "for every".
$\in$	**Set Membership**	Read as "is a member of," "belongs to," or "is in".
$\mathbb{N}$	**Natural Numbers**	Stands for the set of *nonnegative integers*, namely $\{0, 1, 2, 3, \ldots\}$.
(n)	**Function / Predicate**	An expression or value evaluated at n.

Proposition 1.3.2. *$a^4 + b^4 + c^4 = d^4$ has no solution when a, b, c, d are positive integers.*

Euler (pronounced "oiler") conjectured this proposition to be true in 1769. It was checked by humans and then by computers for many values of a, b, c, and d over the next two centuries. Ultimately the proposition was proven false in 1987 by Noam Elkies. The solution he found was $a = 95800, b = 217519, c = 414560, d = 422481$. No wonder it took 218 years to show the proposition is false!

In logical notation, Proposition 1.3.2 could be written,

$$\forall a \in \mathbb{Z}^+ \, \forall b \in \mathbb{Z}^+ \, \forall c \in \mathbb{Z}^+ \, \forall d \in \mathbb{Z}^+ . \, a^4 + b^4 + c^4 \neq d^4.$$

Here, $\mathbb{Z}^+$ is a symbol for the positive integers. Strings of $\forall$'s are usually abbreviated for easier reading, as follows:

$$\forall a, b, c, d \in \mathbb{Z}^+ . \, a^4 + b^4 + c^4 \neq d^4.$$

1.3.2 Predicates

A predicate is a proposition whose truth depends on the value of one or more variables.

use a single letter to detonate

$$P(n) ::= \text{“}n \text{ is a perfect square”}$$

Now P(4) is true, and P(5) is false.

1.3.3 Quantifiers

There are a couple of assertions commonly made about a predicate: *that it is sometimes true* and *that it is always true.*

Always True

For all n, $P(n)$ is true.	For all $x \in \mathbb{R}$, $x^2 \geq 0$.
$P(n)$ is true for every n.	$x^2 \geq 0$ for every $x \in \mathbb{R}$.

Sometimes True

There exists an n such that $P(n)$ is true.	There exists an $x \in \mathbb{R}$ such that $5x^2 - 7 = 0$.
$P(n)$ is true for some n.	$5x^2 - 7 = 0$ for some $x \in \mathbb{R}$.
$P(n)$ is true for at least one n.	$5x^2 - 7 = 0$ for at least one $x \in \mathbb{R}$.

an assertion that a predicate is always true, is called a *universally quantified statement.*

An assertion that a predicate is sometimes true, is called *an existentially quantified statement*

1.3.4 More Notation

Universal Quantification:

$$\forall x \in \mathcal{P}(x)$$

Existential Quantification:

$$x \in \mathcal{P}(x)$$

Symbol	Name / Meaning	read as	Explanation
$\forall$	**Universal Quantifier**	"for all" (an upside-down **A** for "**A**ll").	Indicates that $P(x)$ is true for all values of x in domain $_$
$\exists$	**Existential Quantifier**	"there exists" (a backward **E** for "**E**xists")	Indicates that $P(x)$ is true for at least one value of x in domain $_$

The symbols $\forall$ and $\exists$ are always followed by a variable—typically with an indication of the set the variable ranges over—and then a predicate, as in the two examples above.

1.3.5 Mixing Quantifiers

1.3.5 Mixing Quantifiers

Many mathematical statements involve several quantifiers. For example, *Goldbach's Conjecture* states:

> "Every even integer greater than 2 is the sum of two primes."

Let's write this more verbosely to make the use of quantification clearer:

> For every even integer n greater than 2, there exist primes p and q such that $n = p + q$.

Let Evens be the set of even integers greater than 2, and let Primes be the set of primes. Then we can write Goldbach's Conjecture in logic notation as follows:

$$\underbrace{\forall n \in \text{Evens}.}_{\substack{\text{for every even} \\ \text{integer } n > 2}} \underbrace{\exists p \in \text{Primes}\ \exists q \in \text{Primes}.}_{\substack{\text{there exist primes} \\ p \text{ and } q \text{ such that}}} n = p + q.$$

The proposition can also be written more simply as

$$\forall n \in \text{Evens}.\ \exists p, q \in \text{Primes}.\ p + q = n.$$

1.3.6 Order of Quantifiers

- **Quantifiers of the same type can be swapped** without changing the meaning:
  - $\forall x\, \forall y\, P(x, y) \equiv \forall y\, \forall x\, P(x, y)$
  - $\exists x\, \exists y\, P(x, y) \equiv \exists y\, \exists x\, P(x, y)$

- **Mixed quantifiers CANNOT be swapped** without changing the meaning:
  - $\forall x\, \exists y\, P(x, y) \not\equiv \exists y\, \forall x\, P(x, y)$

How to Remember the Dependency

When reading logical expressions from left to right:

$\forall x\, \exists y\, P(x, y)$: y can depend on x. (Each x can pick its own y).

$\exists y\, \forall x\, P(x, y)$: y is fixed first. One single y must work for all possible x's simultaneously.

1.3.7 Variable over one domain

When every variable in a logical expression belongs to the same non-empty set D, repeating $\in D$ for every quantifier becomes redundant.

Instead, we define D once as the **domain of discourse** (or simply the **domain**) and omit explicit set membership from the quantifiers:

$$\text{Instead of: } \forall x \in D\, \forall y \in D\, Q(x, y)$$

$$\text{We write: } \forall x y.\ Q(x, y)$$

1.3.8 Negating Quantifiers

$$\neg(\forall x, P(x)) \equiv \exists x, \neg P(x)$$

$$\neg(\exists x, P(x)) \equiv \forall x, \neg P(x)$$

1.4 Validity

A propositional formula is called *valid* when it evaluates to **T** no matter what truth values are assigned to the individual propositional variables

explanation

Two logical expressions A and B are **logically equivalent** ($A \equiv B$) if and only if the biconditional statement $A \iff B$ ($A \leftrightarrow B$) is **valid**.

In the text, the Distributive Law states:

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

To prove this law using validity, we check whether the biconditional formula is valid:

$$[P \wedge (Q \vee R)] \iff [(P \wedge Q) \vee (P \wedge R)]$$

use True table

The same idea extends to predicate formulas, but to be valid, a formula now must evaluate to true no matter what values its variables may take over any unspecified domain, and no matter what interpretation a predicate variable may be given. For example, we already observed that the rule for negating a quantifier is captured by the valid assertion (1.5).

Another useful example of a valid assertion is

$$\exists x \forall y.\ P(x, y) \text{ IMPLIES } \forall y \exists x.\ P(x, y). \tag{1.7}$$

Here's an explanation why this is valid:

Let D be the domain for the variables and P_0 be some binary predicate[4] on D. We need to show that if

$$\exists x \in D \; \forall y \in D. \; P_0(x, y) \tag{1.8}$$

holds under this interpretation, then so does

$$\forall y \in D \; \exists x \in D. \; P_0(x, y). \tag{1.9}$$

So suppose (1.8) is true. Then by definition of $\exists$, this means that some element $d_0 \in D$ has the property that

$$\forall y \in D. \, P_0(d_0, y).$$

By definition of $\forall$, this means that

$$P_0(d_0, d)$$

is true for all $d \in D$. So given any $d \in D$, there is an element in D, namely, d_0, such that $P_0(d_0, d)$ is true. But that's exactly what (1.9) means, so we've proved that (1.9) holds under this interpretation, as required.

1.5 Satisfiability

A proposition is **satisfiable** *if some setting of the variables makes the proposition true*

The general problem of deciding whether a proposition is satisfiable is called **SAT**.

approaches to **SAT**

1. to construct a truth table and check whether or not a T ever appears.

2. Patterns of Proof

2.1 The Axiomatic Method

Starting from axioms, establish the truth of many additional propositions by providing “proofs”.

- A proof is a sequence of logical deductions from axioms and previously-proved statements that concludes with the proposition in question
- Important propositions are called theorems.
- A lemma is a preliminary proposition useful for proving later propositions. (*"a stepping-stone theorem"*)
- A corollary is a proposition that follows in just a few logical steps from a lemma or a theorem. (*"immediate consequence."*)

to get a quick launch : we’ll accept all familiar facts from high school math as a huge set of axioms as our foundation

2.1.2 Logical Deductions

Logical deductions or inference rules are used to prove new propositions using previously proved ones.

A key requirement of an inference rule is that it must be sound: any assignment of truth values that makes all the antecedents true must also make the consequent true.

modus ponens

Rule 2.1.1.

$$\frac{P, \quad P \text{ IMPLIES } Q}{Q}$$

- **Antecedents (Top):** The statements *above* the line. These are assumptions or facts that you have already proven to be true.
- **Conclusion / Consequent (Bottom):** The statement *below* the line. This is the new fact you are allowed to claim as true.

so when an inference rule is sound mean, if P->Q and P is true there is no way Q is false

other inference rules

Rule 2.1.2.

$$\frac{P \text{ IMPLIES } Q, \quad Q \text{ IMPLIES } R}{P \text{ IMPLIES } R}$$

Rule 2.1.3.

$$\frac{P \text{ IMPLIES } Q, \quad \text{NOT}(Q)}{\text{NOT}(P)}$$

Rule 2.1.4.

$$\frac{\text{NOT}(P) \text{ IMPLIES NOT}(Q)}{Q \text{ IMPLIES } P}$$

2.1.3 Proof Templates

When you first start doing proofs, looking at a blank page can feel overwhelming because you have complete freedom to write logical deductions any way you want. A proof template acts like a recipe or a standard skeleton that guides you on:

- **How to start** (e.g., writing down "Assume P" for an implication)
- **What steps to take in the middle** (e.g., showing how the assumption leads to the target conclusion)
- **How to finish** (e.g., concluding with your "doohickey" or Q.E.D. mark)

2.2 Proof by Cases

> Breaking a complicated proof into cases and proving each case separately
>
> - a useful and common proof strategy.
> - works in much more general environments than propositions involving Boolean variables.

example : a club

Imagine you are hosting a party with **6 kids** (let's call them Alice, Bob, Charlie, David, Eve, and Frank).

The big rule of the party is this: **Either you can find 3 kids who all know each other (a "club"), OR you can find 3 kids who are total strangers to each other (a "group of strangers").**

How do we prove this always happens? We use **Proof by Cases**!

Step 1: Pick a Kid to be the Boss

Let's pick **Alex** out of the 6 kids. That leaves **5 other kids** at the party.

When it comes to Alex and those 5 other kids, there are only two possibilities for each of them:

- Either they **know** Alex.
- Or they **do not know** Alex.

Because there are 5 kids total, and everyone either knows Alex or doesn't, we can split the world into **two big buckets (Cases)**:

- **Case 1:** At least **3 kids** in that group **know** Alex.
- **Case 2:** At least **3 kids** in that group **do not know** Alex.

*(One of these two buckets **has** to be true, because 5 kids split into two groups means one group must have at least half—which is 3 or more!)*

Step 2: Look Inside Case 1 (At least 3 kids know Alex)

Let's say kids Bob, Charlie, and David are the ones who know Alex.
Now we look at how *they* relate to each other. This gives us two mini-buckets (Subcases):

- **Subcase 1.1: None of them know each other.**
  - Look at Bob, Charlie, and David. If none of them have ever met each other, boom! You just found **3 mutual strangers**. The theorem is proven true for this subcase!
- **Subcase 1.2: Some pair of them *do* know each other.**
  - Let's say Bob and Charlie know each other.
  - Take Bob, Charlie, and **Alex**. Since Bob and Charlie know each other, and *both* of them know Alex, these three kids all know each other! You just found a **club of 3 friends**. The theorem is proven true for this subcase too!

(Since either they don't know each other or they do, Case 1 is totally covered!)

Step 3: Look Inside Case 2 (At least 3 kids do NOT know Alex)

Let's say Eve, Frank, and Grace are the ones who *don't* know Alex. Just like before, we look at how they relate to each other with two mini-buckets:

- **Subcase 2.1: Every pair of them knows each other.**
  - If Eve, Frank, and Grace all know each other, you have a **club of 3 friends**! The theorem wins.
- **Subcase 2.2: Some pair of them do NOT know each other.**
  - Let's say Eve and Frank don't know each other.

- Take Eve, Frank, and **Alex**. Remember, none of these three even know Alex, *and* Eve and Frank don't know each other either. That means none of these three have met! You just found **3 mutual strangers**. The theorem wins again!

The Grand Finale

Since **Case 1** covers every possibility when 3 people know Alex, and **Case 2** covers every possibility when 3 people don't know Alex, and one of those two *always* has to happen... we have successfully proved that **you can never escape having either a club of 3 or a group of 3 strangers** in a group of 6 people!

That's how **proof by cases** works: you chop the problem up into every possible scenario, show that the rule works in *every single scenario*, and walk away victorious!

2.3 Proving an Implication

Propositions of the form "If P, then Q" are called implications. This implication is often rephrased as "P IMPLIES Q" or "P $\implies$ Q".

2.3.1 Method #1: Assume P is true

- An implication is only **false** in one specific situation: when P is True and Q is False.
- If P is **False**, the entire implication ($P \implies Q$) is **automatically true**
- focus entirely on the interesting case: **What happens when P is actually true?**

Hence, in order to prove that P IMPLIES Q:

1. Write, "Assume P."
2. Show that Q logically follows

example

Theorem 2.3.1. If 0 $\leq$ x $\leq$ 2, then $-x^3 + 4x + 1 > 0$

Before we write a proof of this theorem, we have to do some scratchwork to figure out why it is true.

The inequality certainly holds for $x = 0$; then the left side is equal to 1 and $1 > 0$. As x grows, the $4x$ term (which is positive) initially seems to have greater magnitude than $-x^3$ (which is negative). For example, when $x = 1$, we have $4x = 4$, but $-x^3 = -1$. In fact, it looks like $-x^3$ doesn't begin to dominate $4x$ until $x > 2$. So it seems the $-x^3 + 4x$ part should be nonnegative for all x between 0 and 2, which would imply that $-x^3 + 4x + 1$ is positive.

So far, so good. But we still have to replace all those "seems like" phrases with solid, logical arguments. We can get a better handle on the critical $-x^3 + 4x$ part by factoring it, which is not too hard:

$$-x^3 + 4x = x(2 - x)(2 + x)$$

Aha! For x between 0 and 2, all of the terms on the right side are nonnegative. And a product of nonnegative terms is also nonnegative. Let's organize this blizzard of observations into a clean proof.

Proof. Assume $0 \leq x \leq 2$. Then x, $2 - x$, and $2 + x$ are all nonnegative. Therefore, the product of these terms is also nonnegative. Adding 1 to this product gives a positive number, so:

$$x(2 - x)(2 + x) + 1 > 0$$

Multiplying out on the left side proves that:

$$-x^3 + 4x + 1 > 0$$

as claimed. ■

a couple points that apply to all proofs

There are a couple points here that apply to all proofs:

- You'll often need to do some scratchwork while you're trying to figure out the logical steps of a proof. Your scratchwork can be as disorganized as you like—full of dead-ends, strange diagrams, obscene words, whatever. But keep your scratchwork separate from your final proof, which should be clear and concise.

- Proofs typically begin with the word "Proof" and end with some sort of doohickey like □ or ■ or "q.e.d". The only purpose for these conventions is to clarify where proofs begin and end.

Potential Pitfall

Potential Pitfall

For the purpose of proving an implication P IMPLIES Q, it's OK, and typical, to begin by assuming P. But when the proof is over, it's no longer OK to assume that P holds! For example, Theorem 2.3.1 has the form "if P, then Q" with P being "$0 \leq x \leq 2$" and Q being "$-x^3 + 4x + 1 > 0$," and its proof began by assuming that $0 \leq x \leq 2$. But of course this assumption does not always hold. Indeed, if you were going to prove another result using the variable x, it could be disastrous to have a step where you assume that $0 \leq x \leq 2$ just because you assumed it as part of the proof of Theorem 2.3.1.

2.3.2 Method #2: Prove the Contrapositive

Sometime proving contrapositive is easier than proving the implication

1. Write, "We prove the contrapositive:" and then state the contrapositive.
2. Proceed as in Method #1

example

Proof. We prove the contrapositive: if $\sqrt{r}$ is rational, then r is rational.

Assume that $\sqrt{r}$ is rational. Then there exist integers a and b such that:

$$\sqrt{r} = \frac{a}{b}$$

Squaring both sides gives:

$$r = \frac{a^2}{b^2}$$

Since a^2 and b^2 are integers, r is also rational. ■

2.4 Proving an "If and Only If"

2.4.1 Method #1: Prove Each Statement Implies the Other

2.4.1 Method #1: Prove Each Statement Implies the Other

The statement "P IFF Q" is equivalent to the two statements "P IMPLIES Q" and "Q IMPLIES P". So you can prove an "iff" by proving *two* implications:

1. Write, "We prove P implies Q and vice-versa."
2. Write, "First, we show P implies Q." Do this by one of the methods in Section 2.3.
3. Write, "Now, we show Q implies P." Again, do this by one of the methods in Section 2.3.

Choose Method 1 : "Assume P is True" or Method 2: "Prove the Contrapositive" from OpenCourseWare - Mathematics for Computer Science - 2. induction > 2.3 Proving an Implication

2.4.2 Method #2: Construct a Chain of IFFs

The Core Idea

Instead of proving two separate directions ($P \implies Q$ and then $Q \implies P$ like Method #1), you build a **bridge of logical equivalents**:

1. Start with your first statement (P).
2. Show that P is equivalent to a second statement ($P \iff S_1$).
3. Show that S_1 is equivalent to a third statement ($S_1 \iff S_2$).
4. Keep going step-by-step until you finally reach your target statement (Q).

Because every single step is an "iff" ($A \iff B \iff C \iff \ldots \iff Q$), the whole chain guarantees that your starting statement and ending statement are entirely equivalent.

The Two-Step Recipe

1. **State the plan:** Write down, *"We construct a chain of if-and-only-if implications."*
2. **Execute the chain:** Write your propositions with **iff** connecting each step until you get from P all the way to Q.

example

Theorem 2.4.1. *The standard deviation of a sequence of values $x_1, ..., x_n$ is zero iff all the values are equal to the mean*

Here is how the chain flows logically from start to finish:

Step 1: Start with the Statement (P)

You start with the initial claim that the standard deviation formula equals zero:

$$\sqrt{\frac{(x_1-\mu)^2+(x_2-\mu)^2+\cdots+(x_n-\mu)^2}{n}}=0$$

Step 2: Eliminate the Square Root

- **Reasoning:** Zero is the unique number whose square root is zero.
- **Equivalence ($\iff$):** This holds if and only if the fraction inside is zero, which means the numerator itself equals zero:

$$(x_1-\mu)^2+(x_2-\mu)^2+\cdots+(x_n-\mu)^2=0$$

Step 3: Use the Property of Squares

- **Reasoning:** The square of any real number is always nonnegative (≥ 0). Therefore, you have a sum of nonnegative terms adding up to zero.
- **Equivalence ($\iff$):** For a sum of nonnegative numbers to equal zero, **every single term** individually must be zero:

$$\text{Every term } (x_i-\mu)^2 \text{ is equal to } 0$$

Step 4: Solve the Individual Terms (Q)

- **Reasoning:** A squared term $(x_i - \mu)^2$ equals zero if and only if its base $(x_i - \mu)$ is zero.
- **Equivalence ($\iff$):** $x_i - \mu = 0$, which rearranges to $x_i = \mu$.
- **Conclusion:** Therefore, every value in the sequence equals the mean.

By stringing these logical equivalents together with **iff** arrows, you prove the entire theorem in a clean, elegant linear path without having to write two separate proofs!

2.5 Proof by Contradiction (or indirect proof)

Method: In order to prove a proposition P by contradiction:

1. Write, “We use proof by contradiction.”
2. Write, “Suppose P is false.”
3. Deduce something known to be false (a logical contradiction).
4. Write, “This is a contradiction. Therefore, P must be true.”

example

Thm: $\sqrt{2}$ is irrational

Proof: by contradiction

Assume for the purpose of contradiction that

$\sqrt{2}$ is rational

$\sqrt{2} = \frac{a}{b}$ (where a & b are lowest form)

$2 = \frac{a^2}{b^2}$

$2b^2 = a^2$

since a^2 ~~can be~~ multiple of 2,

$\Rightarrow a^2$ is even

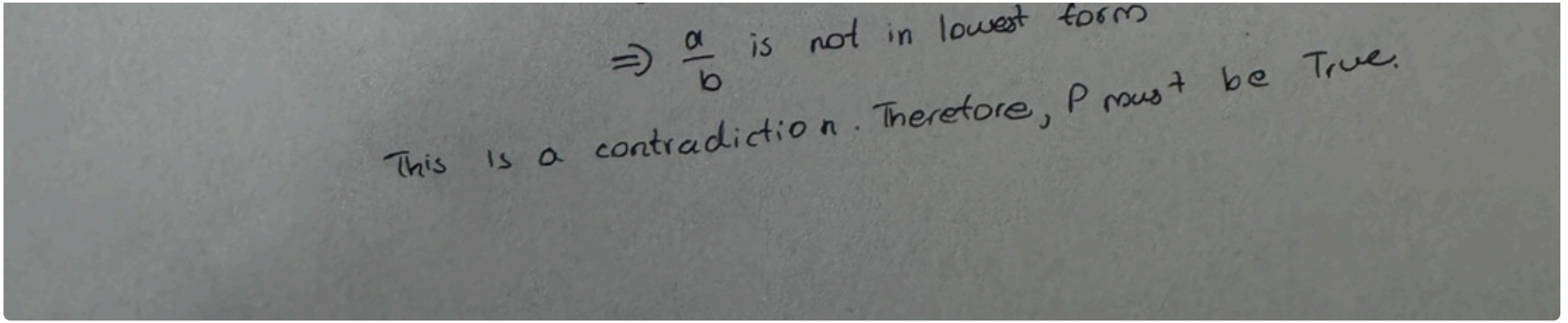
$\Rightarrow a$ is even $(2 \mid a)$

$\Rightarrow 4 \mid a^2$

$\Rightarrow 4 \mid 2b^2$

$\Rightarrow 2 \mid b^2$

$\Rightarrow b$ is even $(2 \mid b)$



Potential Pitfall

Whatever you temporarily assume or discover on the way to a contradiction stays trapped inside that specific proof. Don't let those temporary "what-if" facts leak out into the rest of your math!

2.6 Proofs about Sets

In fact, we have already talked about a lot of sets: the set of integers, the set of real numbers, and the set of positive even numbers, to name a few.

2.6.1 Definitions

a set is a bunch of objects, which are called the elements of the set.

The conventional way to write down a set is to list the elements inside curly-braces. For example, here are some sets:

$$A = \{\text{Alex, Tippy, Shells, Shadow}\} \quad \text{dead pets}$$

$$B = \{\text{red, blue, yellow}\} \quad \text{primary colors}$$

$$C = \{\{a, b\}, \{a, c\}, \{b, c\}\} \quad \text{a set of sets}$$

Other sets might be defined by indicating how to generate a list of them:

$$= \{1, 2, 4, 8, 16, ...\} \quad \text{the powers of 2}$$

- the order of elements is not significant
- duplicate are not allowed

Some popular sets

Some Popular Sets

Mathematicians have devised special symbols to represent some common sets.

symbol	set	elements
$\emptyset$	the empty set	none
$\mathbb{N}$	nonnegative integers	$\{0, 1, 2, 3, \ldots\}$
$\mathbb{Z}$	integers	$\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$
$\mathbb{Q}$	rational numbers	$\frac{1}{2},\ -\frac{5}{3},\ 16,$ etc.
$\mathbb{R}$	real numbers	$\pi,\ e,\ -9,\ \sqrt{2},$ etc.
$\mathbb{C}$	complex numbers	$i,\ \frac{19}{2},\ \sqrt{2} - 2i,$ etc.

A superscript "$+$" restricts a set to its positive elements; for example, $\mathbb{R}^+$ denotes the set of positive real numbers. Similarly, $\mathbb{R}^-$ denotes the set of negative reals.

Comparing and Combining set

comparing

- $S \subseteq T$ -> set S is a subset of set T or equal to set T
- $S \subset T$ -> set S is a subset of set T but never equal to set T

combining

- The *union* of sets X and Y ($X \cup Y$) contains all elements appearing in X or Y.
- The *intersection* of X and Y ($X \cap Y$) consists of all elements that appear in *both* X and Y.
- The *set difference* of X and Y ($X - Y$) consists of all elements that are in X, but not in Y.

The Complement of a Set

the **complement of a set** consists of *all the elements that do **not** belong to that specific set* from same domain

notations : $A', A^c, \quad A, \not{A}$

$A' = \{x \in \not{x}\ \not{A}\}$

Cardinality

The *cardinality* of a set A is the number of elements in A and is denoted by $|A|$. For example,

$$|\emptyset| = 0,$$
$$|\{1, 2, 4\}| = 3, \text{ and}$$
$$|\mathbb{N}| \text{ is infinite.}$$

The power set

so the power set of a set A, is every possible subset of set A

The Power Set

The set of all the subsets of a set, A, is called the *power set*, $\mathcal{P}(A)$, of A. So $B \in \mathcal{P}(A)$ iff $B \subseteq A$. For example, the elements of $\mathcal{P}(\{1,2\})$ are $\emptyset, \{1\}, \{2\}$ and $\{1,2\}$.

More generally, if A has n elements, then there are 2^n sets in $\mathcal{P}(A)$. In other words, if A is finite, then $|\mathcal{P}(A)| = 2^{|A|}$. For this reason, some authors use the notation 2^A instead of $\mathcal{P}(A)$ to denote the power set of A.

Sequences vs sets

like python list vs python set

- The elements of a set are required to be distinct, but terms in a sequence can be the same. Thus, (a, b, a) is a valid sequence of length three, but $\{a, b, a\}$ is a set with two elements—not three.
- The terms in a sequence have a specified order, but the elements of a set do not. For example, (a, b, c) and (a, c, b) are different sequences, but $\{a, b, c\}$ and $\{a, c, b\}$ are the same set.
- Texts differ on notation for the *empty sequence*; we use λ for the empty sequence and $\emptyset$ for the empty set.

Cross Products

The product operation is one link between sets and sequences. A *product of sets*, $S_1 \times S_2 \times \cdots \times S_n$, is a new set consisting of all sequences where the first component is drawn from S_1, the second from S_2, and so forth. For example, $\mathbb{N} \times \{a, b\}$ is the set of all pairs whose first element is a nonnegative integer and whose second element is an a or a b:

$$\mathbb{N} \times \{a, b\} = \{(0, a), (0, b), (1, a), (1, b), (2, a), (2, b), \ldots\}$$

A product of n copies of a set S is denoted S^n. For example, $\{0, 1\}^3$ is the set of all 3-bit sequences:

$$\{0, 1\}^3 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}$$

2.6.2 Set Builder Notation

when we can't describe the elements of a set explicitly, we use *set builder notation* which define *a set using a predicate*; in particular, the set consists of all values that make the predicate true.

$$A ::= \{n \in \mathbb{N} \mid n \text{ is a prime and } n = 4k + 1 \text{ for some integer } k\}$$
$$B ::= \{x \in \mathbb{R} \mid x^3 - 3x + 1 > 0\}$$
$$C ::= \{a + bi \in \mathbb{C} \mid a^2 + 2b^2 \leq 1\}$$

2.6.3 Proving Set Equalities

Two sets are defined to be equal if they contain the same elements.

- $X = Y$ means that $\not\in X$ if and only if $\not\in Y$, for all elements, $\not$

2.7 Good Proofs in Practice

One purpose of a proof is to establish the truth of an assertion with absolute certainty

some general tips on writing good proofs:

State your game plan. A good proof begins by explaining the general line of reasoning. For example, "We use case analysis" or "We argue by contradiction."

Keep a linear flow. Sometimes proofs are written like mathematical mosaics, with juicy tidbits of independent reasoning sprinkled throughout. This is not good. The steps of an argument should follow one another in an intelligible order.

A proof is an essay, not a calculation. Many students initially write proofs the way they compute integrals. The result is a long sequence of expressions without explanation, making it very hard to follow. This is bad. A good proof usually looks like an essay with some equations thrown in. Use complete sentences.

Avoid excessive symbolism. Your reader is probably good at understanding words, but much less skilled at reading arcane mathematical symbols. So use words where you reasonably can.

Revise and simplify. Your readers will be grateful.

Introduce notation thoughtfully. Sometimes an argument can be greatly simplified by introducing a variable, devising a special notation, or defining a new term. But do this sparingly since you're requiring the reader to remember all that new stuff. And remember to actually *define* the meanings of new variables, terms, or notations; don't just start using them!

Structure long proofs. Long programs are usually broken into a hierarchy of smaller

Structure long proofs. Long programs are usually broken into a hierarchy of smaller procedures. Long proofs are much the same. Facts needed in your proof that are easily stated, but not readily proved are best pulled out and proved in preliminary lemmas. Also, if you are repeating essentially the same argument over and over, try to capture that argument in a general lemma, which you can cite repeatedly instead.

Be wary of the "obvious". When familiar or truly obvious facts are needed in a proof, it's OK to label them as such and to not prove them. But remember that what's obvious to you, may not be—and typically is not—obvious to your reader.

Most especially, don't use phrases like "clearly" or "obviously" in an attempt to bully the reader into accepting something you're having trouble proving. Also, go on the alert whenever you see one of these phrases in someone else's proof.

Finish. At some point in a proof, you'll have established all the essential facts you need. Resist the temptation to quit and leave the reader to draw the "obvious" conclusion. Instead, tie everything together yourself and explain why the original claim follows.

In computer science, it is easy to write code that *mostly* works or passes a few simple tests, relying on loose assumptions ("hand-waving") rather than rock-solid logic. But as the text warns, when systems are built on sloppy reasoning instead of rigorous verification, the consequences aren't just minor bugs—they can be **catastrophic**.

Someday, real people will be at the mercy of critical systems that *you* and your classmates design.